Databases 2.0

5th September 2024

Marking Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

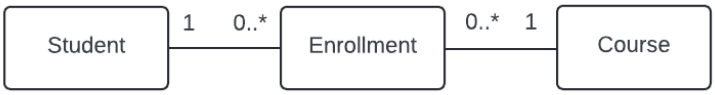
Throughout the marking, please credit any valid alternative point.

Answer ALL questions

Marks

Question 1

A section of a university database system that manages student records and course enrolments is shown below.



a) Define what a primary key is. Provide ONE (1) example of a primary key **and** explain its significance in the context of the university database. **4**

Mark scheme:

A primary key is a unique identifier (1 mark) for each record in a database table (1 mark).

An example of a primary key is student ID (1 mark) in the student table (1 mark).

Any suitable answers should be accepted. Examples should relate to the scenario.

b) Define what a foreign key is. Provide ONE (1) example of a foreign key **and** explain its role in the context of the university database. **4**

Mark scheme:
A foreign key is the copy of an attribute from another entity (1 mark) and links one table to another (1 mark), maintaining referential integrity (1 mark).

An example of a foreign key is course ID in the enrolment table links to course ID in the course table (1 mark), ensuring valid course assignments (1 mark)

A maximum of 2 marks for correctly defining a foreign key.
1 mark for a suitable example and 1 mark for its role, up to a maximum of 2 marks.

Any suitable answers should be accepted. Examples should relate to the scenario.

c) Explain TWO (2) advantages of normalisation. **2**
Mark scheme:
Reduce data redundancy (1 mark), enhance data accuracy (1 mark) and improve data integrity (1 mark).
A maximum of 2 marks.

Total 10 Marks

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Marks

Question 2

10

Identify **and** explain FIVE (5) functions a Relational DBMS performs.

Mark scheme:

Award 1 mark for identifying a correct point, and 1 mark for explanation, up to a maximum of 10 marks.

Organises data into tables consisting of rows and columns, (1 mark) which are related to each other. (1 mark)

Allows efficient access to data (1 mark)

Ensures there is no redundant data (1 mark)

CRUD (1 mark): Stands for Create, Retrieve, Update and Delete (1 mark). The basic interactions between application and the data (1 mark).

Data dictionary (1 mark): Stores definitions of the data elements and their relationships in a data dictionary: Metadata, Table structures, PKs, data types, constraints and etc (1 mark).

Transaction management (1 mark): A transaction is one or more operations that access or make a change to the database. (1 mark)

Concurrency control (1 mark): The ability for many users to perform transactions at the same time. (1 mark)

Recovery (1 mark): Database must be capable of being recovered in the event of a hardware or software failure (1 mark).

Authorisation (1 mark): Allocate different roles and levels of access to users with associated usernames, passwords and privileges that give access to some, but not all areas of the database (1 mark), or award 1 mark for referring to security.

Data communication (1 mark): Interact with other software (1 mark) enabling connection between the kernel and the toolkit software. (1 mark)

Administration utilities (1 mark): allow importing and exporting of data (1 mark), monitoring of use and monitoring of performance (1 mark).

Data integrity (1 mark): make sure the data accurately reflects the model (1 mark). This involves the use of integrity constraints (1 mark).

Total 10 marks

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Marks

Question 3

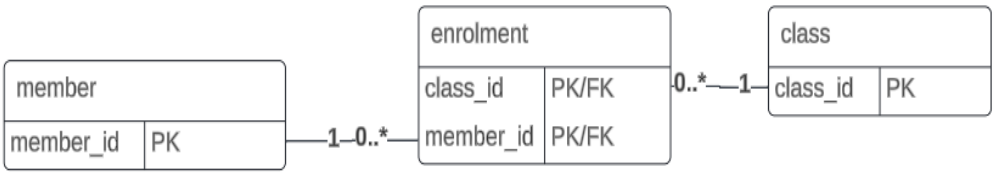
10

Scenario:

A fitness centre offers various classes such as yoga, spinning and boxing. The system tracks which members are enrolled in which classes.

Draw an Entity Relationship Diagram to represent the fitness centre's class management system described above. You need to identify entities, relationships, cardinalities and label all the primary and foreign keys. Clearly state which entity the keys belong to.

Mark scheme:



1 mark for each entity, up to a maximum of 3 marks. Entity names may vary.

1 mark for both relationships, and 1 mark each for cardinalities. The notation types should not be the focus of this marking.

1 mark for each correctly labelled PK and FK, up to a maximum of 4 marks. An enrolment_id as PK should be accepted but must be along with class_id and member_id as FK.

Total 10 marks

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Marks

Question 4

a) A car rental company manages its fleet of cars, customer reservations and maintenance records. Each car can be rented by multiple customers over time and each customer can make multiple reservations. Maintenance records track the service history of each car. **4**

Entities:

- Car
- Customer
- Reservation
- Maintenance

Create a CRUD matrix to show the following transactions:

Transaction 1 – Add a new car to the fleet

Transaction 2 – Create a new reservation for an existing customer

Transaction 3 – Remove a car from the fleet and all its maintenance records

Transaction 4 – Update a car's maintenance record after a service

Mark scheme:

Award 1 mark for each correct row in a table, up to a maximum of 4 marks. 0.5 mark should be given for a partially correct row. Marks should be rounded up at the end of the question.

Transaction	*Car*	*Customer*	*Reservation*	*Maintenance*
T1	*C*			
T2		*R*	*C*	
T3	*D*			*D*
T4	*R*			*U*

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b) A number of business rules have been defined for the car rental company's database system. Discuss how each might be enforced on the system: **6**

Rule 1: All cars must have a unique identification number.
Rule 2: Reservations cannot be made for cars that are currently under maintenance.
Rule 3: Each maintenance record must include the date of service and the details of the service performed.

Mark scheme:

2 marks for each accurately description of the database mechanisms for enforcing the business rules, up to a maximum of 6 marks.

Rule 1: Implement a unique constrainst on the car_id/car_reg (name can vary) field in the car table.

Rule 2: Add a status attribute to the car table (e.g., available, under maintenance) and enforce a check constraint to prevent reservation creation for cars marked as not available/under maintenance.

Rule 3: Enforce NOT NULL constraints on the service date and service details fields in the maintenance table to ensure that every maintenance record includes the necessary information.

Total 10 marks

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Marks

Question 5

a) A local library maintains a spreadsheet to track book borrowings, recording each borrower's ID and name along with the titles and due dates for books borrowed. Each borrower can borrow multiple books, but the library holds only one copy of each book. The current structure of the data does not conform to the First Normal Form. Convert the provided data into 1NF. Make sure you clearly identify keys. **8**

Provided data:

borrower_id	borrower_name	borrowed_books
101	Bob Smith	"The Pragmatic Programmer" due 2024-10-01 "Intro to DB Systems" due 2024-10-08
102	Alice Brown	"Python Crash Course" due 2024-12-15
103	Jane Doe	"The Art of Computer Programming" due 2024-09-23 "Data Science for Dummies" due 2024-10-31

Mark scheme:

borrower_id(PK)	***first_name***	***last_name***
101	***Bob***	***Smith***
102	***Alice***	***Brown***
103	***Jane***	***Doe***

book_id(PK)	***book_title***	***due_date***	***borrower_id (FK)***
1901	***The Pragmatic Programmer***	***2024-10-01***	***101***
1902	***Python Crash Course***	***2024-12-15***	***102***
1903	***The Art of Computer Programming***	***2024-09-23***	***103***
1904	***Intro to DB Systems***	***2024-10-08***	***101***

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1 mark for dividing the borrower_name column into two separate columns: first_name and last_name.

1 mark for each correctly identified table, up to a maximum of 2 marks.

1 mark for including borrower_id in the book table, and 1 mark if it's identified as an FK.

1 mark for the presence of book_title and due_date in the book table.
1 mark for including book_id in the book table, and 1 mark if it's identified as a PK.

b) A small business has been using flat files to manage its data but is considering switching to a database system. Identify TWO (2) advantages of using a database system over flat file. **2**

Mark scheme:

1 mark for each correct advantage, up to 2 marks. Suggested answers are not limited to:

- ***Data integrity***
- ***Data consistency***
- ***Efficient data retrieval***
- ***Automatic report generation***
- ***Scalability/handle large amount of data***
- ***Concurrent access by multiple users***
- ***Backup/security controls***

Any other suitable answers should be accepted.

Total 10 marks

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Marks

Question 6

Consider the data in the following table. Write the appropriate SQL command to perform each of the following tasks.

movie

movie_id	movie_title	rental_price	max_rental_days
1	Inception	8.99	5
2	Toy Story	5.99	7
3	Avatar	7.99	6

a) Create the `movie` table. **6**

Mark scheme:

CREATE TABLE movie (

movie_id, INT NOT NULL,

movie_title VARCHAR(100),

rental_price DECIMAL,

max_rental_days INT,

PRIMARY KEY (movie_id));

1 mark for each correctly defined row in the SQL command, up to a maximum of 6 marks.

b) Update the `movie` table to include a new column named `rental_date` to record the date of the last rental. **2**

Mark scheme:

ALTER TABLE movie
ADD rental_date DATE/TIME;
1 mark for each correctly defined row in the SQL command, up to 2 marks.

c) Remove the `max_rental_days` column in the movie table. **2**

Mark scheme:

ALTER TABLE movie
DROP COLUMN max_rental_days;
1 mark for each correctly defined row in the SQL command, up to 2 marks.

Total 10 marks

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Marks

Question 7

Scenario: Technology Retailer Database

A technology retailer has set up a database to manage information about its products, customers, and orders. Below are the tables with sample data.

Product

product_id	product_name	category_id
1	laptop	1
2	smartphone	2
3	headphones	3
4	monitor	1

Category

category_id	category_name
1	Computers
2	Mobile Devices
3	Accessories

Order

order_id	product_id	order_date
1	1	2024-06-23
2	2	2024-07-02
3	3	2024-09-14
4	2	2024-09-15
5	4	2024-09-16

a) Write an SQL command to retrieve a list of all products along with their category names. Include both the product's name and the category name in your results. **3**

Mark scheme:

SELECT p.product_name, c.category_name

FROM product p

JOIN cateogry c ON p.category_id = c.category_id;

1 mark for each correctly defined row in the SQL command, up to a maximum of 3 marks.

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Alternative:

```
SELECT Product.product_name, Category.category_name
 FROM Product, Category
WHERE Category.category_id = Product.category_id;
```

b) Write an SQL command to find the total number of orders placed for each product. The result should list the product name and the total number of orders. **3**

Mark scheme:

SELECT p.product_name, COUNT(o.product_id) AS total_orders (1 mark)

FROM order o

JOIN product p ON o.product_id = p.product_id (1 mark)

GROUP BY p.product_name; (1 mark)

c) Write an SQL command to list the names of the products that have been ordered more than once. Include the product names and the number of times they have been ordered in your results. **4**

Mark scheme:

SELECT p.product_name, COUNT(o.order_id) AS order_count (1 mark)

FROM product p

JOIN order o ON p.product_id = o.product_id (1 mark)
GROUP BY p.product_name (1 mark)

HAVING COUNT(o.order_id) > 1; (1 mark)

Total 10 marks

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Marks

Question 8

A small restaurant has been using manual data recording system to manage its inventory and customer orders. The restaurant tracks information such as ingredient stock levels, supplier details and daily orders in physical logbooks. The restaurant is considering switching to a database system to improve data management. **10**

Identify FIVE (5) advantages of using a database system over a manual system. Provide ONE (1) specific example to illustrate each advantage.

Mark scheme:

Suggested answers are not limited to:

- ***Improve data accuracy***
- ***Efficient data processing/retrieval***
- ***Automatic calculation***
- ***Better data security***
- ***Handle large amount of data efficiently***
- ***Backup***
- ***Automated report generation***
- ***Natural disasters recovery***

1 mark for each correctly identified advantage and 1 mark for each example, up to a maximum of 10 marks.

Any suitable advantages and examples should be accepted.

Total 10 marks

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Marks

Question 9

a) A medium-sized online retail company has experienced several security incidents recently involving their database. **6**
Identify THREE (3) areas of loss that the company might face due to inadequate database security. Provide a brief explanation of each area of loss.

Mark scheme:

Suggested answers are not limited to:

- ***Confidentiality loss: Breaches in database security can lead to the exposure of confidential business information.***
- ***Integrity loss: accidental or malicious modifications to the database can compromise the accuracy and reliability of data.***
- ***Availability loss: service outages or system failures can make the database unavailable.***
- ***Legal and regulatory penalties: failure to protect data adequately can result in non-compliance with data protection regulations.***
- ***Privacy loss: the exposure of personal customer/employee information, violating their privacy.***
- ***Financial loss: direct financial losses due to fraud, legal penalties, and compensation to affected customers.***
- ***Operational disruptions: security incidents can disrupt business operations, causing delays in order processing and delivery.***
- ***Reputation***
- ***Employee productivity***

1 mark for each correctly identified area, up to a maximum of 3 marks.

1 mark for each correct description/ elaboration, up to a maximum of 3 marks.
Any other suitable answers should be accepted.

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b) Identify TWO (2) models of Cloud databases and describe each model. **4**

Mark scheme:

- ***Self-managed database: offers full control over database software and infrastructure. Require in-house expertise for setup, maintenance, and troubleshooting. The responsibility for setup, configuration, management, maintenance, and optimisation lies with organisations/individuals using the database. Suited for organisation with specific database management requirements that cloud solutions cannot meet.***

- ***Managed: cloud providers manage the infrastructure, database software, and administrative tasks. Offers automated backups, updates, and patch management. Reduce the need for in-house database management expertise. Allow organisations to focus on development and application optimisation.***

1 mark for each correctly identified model, up to a maximum of 2 marks.

1 mark for each correct description, up to a maximum of 2 marks.

Any suitable description should be accepted.

Total 10 marks

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Marks

Question 10

Data models are abstract representations of the data structures and the relationships between data elements. Identify **and** explain FIVE (5) different data models. **10**

Mark scheme:

- ***Relational data model: tables (relations) of rows and columns. Tables can have PK and FK to establish relationships between tables.***
- ***Document data model: data stored in JSON or XML documents flexible scheme for semi-structured data like web content, logs (1 mark). Useful for content management systems and e-commerce applications.***
- ***Key-value data model: fast data access through direct key lookup. Useful for user profiles.***
- ***Graph data model: Data represented as nodes and edges. Suitable for highly interconnected data. Useful for social media networking.***
- ***Wide column data model: data stored in tables, rows and dynamic columns. Columns don't need to be defined before data storage. Names and formats of the columns can vary across rows. Quickly accesses, writes, and aggregates data. Handles large volumes of data with complex relationships.***

1 mark for each correctly identified data model, up to a maximum of 5 marks.

1 mark for each correct explanation, up to a maximum of 5 marks.

Any other suitable answers should be accepted.

Total 10 marks

End of Paper

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Learning Outcomes matrix

Question	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	1, 2	Yes
2	1	Yes
3	3	Yes
4	1, 4, 5	Yes
5	2, 4	Yes
6	5	Yes
7	5	Yes
8	1	Yes
9	1, 7	Yes
10	6	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Understand the concepts associated with database systems	Has little to no awareness of principles and concepts underlying theoretical frameworks and approaches and is unable to identify associated strengths or weaknesses.	Has vague awareness of principles and concepts underlying theoretical frameworks and approaches and is only superficially able to identify associated strengths or weaknesses.	Has satisfactory awareness of principles and concepts underlying theoretical frameworks and approaches and demonstrates sufficient ability to identify associated strengths and weaknesses.	Has very good awareness of principles and concepts underlying theoretical frameworks and approaches and is able in detail to identify associated strengths and weaknesses.	Has an excellent awareness of principles and concepts underlying theoretical frameworks and approaches and is comprehensively able to identify associated strengths and weaknesses.

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Understand the concepts associated with the relational model	Has little to no awareness of principles and concepts underlying theoretical frameworks and approaches and is unable to identify associated strengths or weaknesses.	Has vague awareness of principles and concepts underlying theoretical frameworks and approaches and is only superficially able to identify associated strengths or weaknesses.	Has satisfactory awareness of principles and concepts underlying theoretical frameworks and approaches and demonstrates sufficient ability to identify associated strengths and weaknesses.	Has very good awareness of principles and concepts underlying theoretical frameworks and approaches and is able in detail to identify associated strengths and weaknesses.	Has an excellent awareness of principles and concepts underlying theoretical frameworks and approaches and is comprehensively able to identify associated strengths and weaknesses.
Understand how to design and develop a database system	Demonstrates little to no ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an extremely limited ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a satisfactory ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a very good ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an excellent ability to use design principles to effectively develop an artefact to solve an identified issue.
Be able to develop and enhance a logical database design	Demonstrates little to no ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an extremely limited ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a satisfactory ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a very good ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an excellent ability to use design principles to effectively develop an artefact to solve an identified issue.

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Be able to develop and enhance a database system using SQL	Demonstrates little to no ability to adequately develop an artefact to solve an identified issue.	Demonstrates an extremely limited ability to adequately develop an artefact to solve an identified issue.	Demonstrates a satisfactory ability to develop an artefact to solve an identified issue.	Demonstrates a very good ability to develop an artefact to solve an identified issue.	Demonstrates an excellent ability to develop an artefact to solve an identified issue.
Understand the principles behind Big Data and the NoSQL databases	Displays no understanding of Big Data and NoSQL databases. Unable to recognise the key features and benefits of NoSQL over traditional databases or how Big Data impacts database design and scalability.	Shows a basic awareness of Big Data and NoSQL databases. Recognises some key aspects but fails to understand the broader implications and technical considerations involved.	Demonstrates satisfactory comprehension of Big Data principles and the operational basics of NoSQL databases. Can distinguish NoSQL databases from traditional databases and understands their use cases.	Exhibits a strong grasp of the principles behind Big Data and NoSQL databases. Can articulate the advantages of NoSQL databases in handling large datasets and different data types.	Has an in-depth understanding of Big Data and NoSQL databases. Can critically evaluate and recommend NoSQL database solutions for complex Big Data problems, demonstrating a sophisticated understanding of scalability and performance optimisation.
Understand the concepts and principles of Cloud database	Lacks understanding of the fundamental concepts and principles of Cloud databases. Cannot identify the main advantages or the operational model of Cloud databases.	Has a superficial grasp of Cloud databases. Recognises some concepts but cannot articulate how Cloud databases differ from on-premises solutions or their benefits.	Has a solid understanding of Cloud database concepts and principles. Understands the benefits of scalability, availability, and managed services provided by Cloud databases.	Possesses a very good understanding of Cloud databases. Can discuss the importance of Cloud databases in modern architectures and understands their significance in providing flexible and scalable data	Demonstrates an excellent and comprehensive understanding of the key concepts and principles of Cloud databases. Can describe in detail how Cloud databases offer advantages in terms of accessibility, maintenance, and scalability, and can suggest

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				storage solutions.	appropriate use cases for Cloud databases.

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